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College of Health Sciences School of Pharmacy

Chemistry lab report

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1.

Chemistry lab report

Date:jan,11,2021 G.C

EXPERIMENT NUMBER : 1, Recrystallization

THEORY : the purification of solids by crystallization is based upon differences in their solubility in a given solvent.in its simplest form the crystallization process consists of ;

A, dissolving the impure substance in some suitable solvent at or near the boiling point and filter the solution while it is still hot.

B, allowing the hot solution to cool thus causing the dissolved substance to crystallize out and

C, separating the crystals from the solution

APPARATUS AND MATERIALS

- Erlenmeyer flask Suction/filter flask
- Stemless funnel
- Measuring cylinder
- Filter paper
- Buchner funnel
- Suction/filter flask
- Glass rod
- Iron stand
- Clamp
- Iron ring
- Paper box
- Fluted paper
- Water
- Benzoic acid
- Water aspirator
- Ice

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PROCEDURE:

- 1) We weighed 2g of impure benzoic acid and placed it in Erlenmeyer flask, then we poured 100ml of water.
- 2) We fixed the clamp on the neck of Erlenmeyer flask and heated it on stove. we made a fluted paper and put it on the gravity filtration set up.
- 3) After the solution boiled, we started filtering the solution while it was hot by small portion until finished.
- 4) We waited until the solution become cool at room temperature, then put it under ice.
- 5) We prepared suction filtration set up and connected it with water aspirator. We measured the mass of filter paper, which was 1.699g, and placed it on Buchner funnel.
- 6) We turned the tap water and filtered the crystals by pouring all of the contents of the Erlenmeyer flask into the Buchner funnel.
- 7) We prepared a paper box and put the benzoic acid that was filtered on Buchner funnel with the filter paper.
- 8) After 2 weeks, we measured the mass of benzoic acid and the filter paper and it was 4.07g.

RESULT AND DISCUSSION:

We found the mass of benzoic acid by subtracting the mass of filter paper from the total mass.

Mass of benzoic acid = total mass of both benzoic acid and filter paper-mass of filter paper

Mass of benzoic acid = 4.07g – 1.699g

Mass of benzoic acid= 2.371g

- ✓ The percentage impurity of our sample can't be calculated because of the error of balance.

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QUIZ QUESTIONS AND THEIR ANSWERS;

1)How does a water aspirator create partial vacuum?

- ✓ When water from the tap is forced at a high velocity through a small nozzle, at the smallest side of the nozzle a vacuum is created.

2)what is the significance of the term “like dissolves like”?

- ✓ The term is useful for us to choose the right solvent.

3)what are the difference between an ordinary paper and filter paper?

- ✓ The filter paper is relatively strong and it has a tiny pores that allows fluid pass through it. Since it is made of fibers.

5)some solids make perfect crystals, while others are amorphous. what may be the cause for this?

- ✓ The cause for some substances to be amorphous and others to be crystalline is the atomic arrangement difference. as for crystalline they are arranged in definite and repeating 3D patterns in highly ordered microscopic structure.

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Date;jan,25,2021G.C

EXPERIMENT NUMBER;2 ; Determination of melting point

THEORY; The melting point of an organic compound is determined by introducing a small amount of substance in a capillary tube which is then attached to a thermometer partially immersed in an oil bath. The oil heated slowly till the solid substance completely melts. one should carefully note the temperature at which melting starts and continue to observe in order to be able to note the temperature at which melting process ends. melting point is reported as a range. a pure sample has a sharp melting point with very small range, while an impure substance has a lower melting point with a broad range.

APPARATUS AND MATERIALS;

- ❖ Thermometer
- ❖ Rubber band
- ❖ Capillary tube
- ❖ Test tube
- ❖ Bunsen burner
- ❖ Glass tube
- ❖ Iron Stand
- ❖ Benzoic acid from the first experiment
- ❖ Oil
- ❖ Bosshead
- ❖ cork

PROCEDURE;

1. Test tube was attached with iron stand by using boss head.
2. Oil was poured into the test tube
3. Thermometer was fixed across a cork. then rubber band was attached at the bottom.
4. Sample of benzoic acid was added into a capillary tube with the help of glass tube. Then attached it with a thermometer by the rubber band.
5. The cork covered the mouth of the test tube the thermometer and bottom of the capillary tube was dipped into the oil inside.
6. Bunsen burner was lighted and start to heat the test tube. in order to heat the oil evenly, Bunsen burner was dangled.
7. Then melting started at 121°C and melting ended at 123°C.

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RESULT AND DISCUSSION ;

The melting point of benzoic acid was found in the range of 121 °C to 123 °C.

Since the range is only 2, we concluded that the benzoic acid we used was pure. therefore, the recrystallization experiment we performed previously was successful, so is our test for melting point of benzoic acid since the exact melting point range of it is 121 °C to 122 °C.

QUIZ QUESTIONS AND THEIR ANSWERS

1) Why is melting point often reported as range and not as exact figure?

- ✓ This is done due to changes to the physical properties of some compounds at different atmospheric conditions. a basic example is nothing that water boils more quickly at a higher altitude.
- ✓ The measuring instrument and observer also matters. No matter how expert we are or how accurate is our device there are always chances of miscalculation or estimation.

2) Why does an impurity reduce the melting point of a substance?

- ✓ Foreign substances in a crystalline solid disrupt the repeating pattern of forces that holds the solid together. therefore a smaller amount of energy is required to melt the part of the solid surrounding the impurity. this explains the melting point depression(lowering) observed from impure solids.

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DARE;Jan,25,2021G.C

EXPERIMENT; 3 ; Simple distillation

THEORY; Distillation is one of the most common methods of purifying liquids. if a liquid sample is contaminated with impurities that are non volatile, or if it is contaminated with another liquid which boils at least 50°c higher than the boiling point of the desired liquid, the main liquid can be obtained in a pure form by simple distillation. thus, simple distillation is useful when only a small amount of impurity is present or when the liquid to be separated have widely differing boiling points.

Simple distillation involves refining a liquid through the process of separation. the most common purpose of simple distillation is to purify drinking water of unwanted chemicals and minerals such as salt. most commonly it is used in the production of alcoholic beverages such as vodka and whiskey.

APPARATUS AND MATERIALS;

- Distillation flask
- Cork
- Condenser
- Bunsen burner
- Iron stand
- Thermometer
- Measuring cylinders
- Boiling chips
- water
- Contaminated ethanol
- Iron bowl
- Iron ring
- Clamp
- Bosshead
- Thin plastic tube

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PROCEDURE;

- 1) 2 iron stands were put near to each other. the one was fixed with iron ring by bosshead. Iron bowl was put on the ring. Above the iron ring clamp fixed by using bosshead putting the distillation flask partially in to the iron bowl.
- 2) Condenser was attached with the distillation flask by plastic cork. other end of condenser was attached with another iron stand by clamp which was attached by bosshead to the stand.
- 3) 2 thin plastic tubes were attached with the 2 outlets of the condenser. The other end of upper attached tube fixed with water pipe. and the other tube end left as an outlet.
- 4) 25 ml of contaminated ethanol was poured into the distillation flask with 3 boiling chips and then covered with a cork having thermometer at the middle.
- 5) Water was poured into iron bowl and measuring cylinder was placed at the bottom of condenser lower end.
- 6) We ignited the Bunsen burner and placed it under the iron bowl.
- 7) Then the water started boiling, first condensed drop of ethanol was observed at 68°C. the process proceeds until the last drop of ethanol was dipped at 74°C.
- 8) Finally 22ml of pure ethanol was found in the measuring cylinder.

RESULT AND DISCUSSION;

25 ml of impure ethanol was added into the distillation flask. after distillation 22ml of pure ethanol was found. this implies that 3 ml was the impurity. Ethanol and the impurity have highly deviating boiling points minimum up to 50°C. it was observed that the boiling point of ethanol was reduced by the impurity from 78.5°C to 68°C. and final temperature was 74°C.

The impurities were left in the distillation flask having a deep red colour.

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QUIZ QUESTIONS AND THEIR ANSWERS;

- 1) Why doesn't all the liquid vaporize at once when the boiling point is reached?
 - Because of boiling point difference in our case the impurities have higher boiling point than that of the boiling point reached.
- 2) An organic liquid begins to decompose at 80°C. its vapor pressure at that temperature is 36mm. how would you distill the liquid?
 - We use a zeotropic distillation in which we add toluene (boiling at 110.6°C) to the substance.
- 3) Why is cold water circulated through a condenser from the bottom upward rather than from the top to downward?
 - The water is circulated from bottom to top to have even cooling. the condenser at the upper part will be hotter than the bottom as we go down it will be colder and colder. when the colder water moves up it will be hotter than the bottom. therefore collision of hot and cold will be avoided. so, the upper part will be cooled as the cold water passes continuously. this also helps to condense the vaporized liquid.
- 4) If the liquid to be distilled had a boiling point higher than 90°C. what kind of bath will you use?
 - We would use alternative baths such as oil bath, silicon bath or sand bath.
- 5) Why is it that when a pure liquid is boiled its temperature remains constant?
 - The temperature of pure substance remains constant during its melting and boiling point. because the change in any state of matter as solid to liquid or the liquid to a gas involves crossing the latent heat of fusion, which causes a change in intermolecular spacing of the molecule.

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Date;feb,01,2021 G.C

EXPERIMENT ; 6;Survey of some functional groups

THEORY; a functional group is a unique combination of atoms present in a molecule. That is particularly responsible for the characteristic set of reactions that the molecule exhibits. Some of the most important functional groups include hydroxyl, olefin, ketone, aldehyde, amine, carboxylic acid, ester, acid halide, anhydride, nitrile, ether, halide, aromatic ring etc. It is possible to distinguish by means of simple tests one functional group from the other.

Alkanes, the simplest of all organic compounds lack the typical functional groups making them most unreactive. They can be inert even in the presence of concentrated sulphuric acid. Alkanes react slowly or not at all with bromine in the dark. However, in the presence of light they react fairly rapidly. This can be explained by means of radical chemistry.

The simplest of alkanes is methane, which is a gas at room temperature, so are ethane, propane, and butane. Pentane and hexane are low boiling liquids and are among the most useful solvents in laboratory as well as in industry. Mixtures of alkanes obtained during fractional distillation of petroleum are called petroleum ether or simply petrol. They are usually categorized by means of their boiling point range. A low boiling petroleum ether has boiling point of 40-60°C, while the higher boiling ones are 60-80°C and 80-120°C.

Higher boiling Naphta or kerosene is the most common sources of energy. This fuel is mixture of alkanes such as isomeric pentanes, hexanes, heptanes, and octanes and small amounts of unsaturated and aromatic hydrocarbons.

APPARATUS AND MATERIALS;

- ❖ Small test tubes
- ❖ Litmus paper
- ❖ Droppers
- ❖ N-hexane
- ❖ Sulfuric acid
- ❖ Water
- ❖ Ethanol
- ❖ Petroleum ether
- ❖ Solution of bromine in carbon tetra chloride or bromine water
- ❖ Cyclohexene
- ❖ Potassium permanganate
- ❖ Benzene
- ❖ Toluene
- ❖ Ice
- ❖ Nitric acid
- ❖ DNPH

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- ❖ Acetone
- ❖ Phenol
- ❖ Ferric chloride solution
- ❖ Kerosene
- ❖ Calcium carbide

PROCEDURE;

1) ALKANES

- i. To test solubility of alkanes 4 small test tubes were used and N-Hexane was poured into all test tubes. Then we added 5 drops of water, ethanol, petroleum ether and sulfuric acid in each test tubes respectively.
- ii. In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. 4-5 drops of solution of bromine in carbon tetra chloride was added into each. then place in dark for 15 minutes. blue litmus paper was fixed on the mouth of each test tubes to test reaction of alkanes in the dark.
- iii. In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. 4-5 drops of solution of bromine in carbon tetra chloride was added into each. then place in light for 5 minutes. blue litmus paper was fixed on the mouth of each test tubes to test solubility reaction in light.
- iv. In separate test tubes 1ml portion of petroleum ether, N-Hexane and kerosene were poured. then 1ml of potassium permanganate was added in each test tube to test solubility.

2) ALKENES

- I. 0.5 ml of cyclohexene was poured into a test tube. then 3 drops of sulfuric acid was added into 1 test tube to test solubility.
- II. 0.5 ml of cyclohexene was poured into a test tube. then 3 drops of bromine water was added to test if they react.
- III. 0.5 ml of cyclohexene was poured into a test tube. then 1 ml of potassium permanganate was added into it to test reaction.

3) ALKYNES

Acetylene generator was set up by our instructor in the fume hood. then water drop was allowed drop by drop into a calcium carbide to test if acetylene was generated or not. bromination test (with bromine water) and baeyer's test (with potassium permanganate).

4) AROMATIC HYDROCARBONS

- a. 6 drops of benzene were added in the test tube then 3 drops of bromine water was added to test the reaction of unsaturated.
- b. In test tube nitrating mixture by adding 10 drops of concentrated sulfuric acid, 3 drops of concentrated nitric acid and 4 drops of toluene together. then we shook the mixture and put it for 15 minutes.

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5) KETONE

10 drops of 2,4-dinitrophenyl-hydrazine(DNPH) was placed in a test tube. then, 10 drops of acetone poured to observe if precipitate was formed.

6) PHENOL

20 drops of phenol was poured to a test tube and 3 drops of neutral ferric chloride solution was added into that test tube.

RESULT AND DISSCUSSION:

Note that

- “n” represents no reaction for reaction test and insoluble for solubility test.
- “y” represents there exist a reaction for reaction test and soluble for solubility test.

1)ALKANE

i.

Alkane	water	ethanol	Petroleum ether	Sulfuric acid
N-Hexane	n	y	y	n
kerosene	N	y	y	n

Alkane	i. Dark reaction	ii. Light reaction	Effect of KMnO_4
Petroleum ether	n and no change of litmus paper also color of solution	Y and blue litmus turned red also solution decolorized	N
N-Hexane	n and no change of litmus paper also color of solution	n and no change of litmus ,solution slightly faded	N
Kerosene	n and no change of litmus paper also color of solution	Y and blue litmus turned red also solution decolorized	N

2)ALKENE

Alkene	Sulfuric acid	Bromine water	KMnO_4
cyclohexene	y	y	y

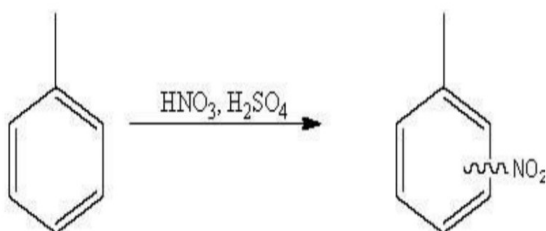
3)ALKYNE

When calcium carbide reacted with water, acetylene was generated and was proved that it existed because when the gas outlet put in bromine water and potassium permanganate decolorized both. this indicates reaction took place between them. and also the presence of acetylene was seen when match lighted near it created acetylene welding flame.

4)AROMATIC HYDROCARBONS

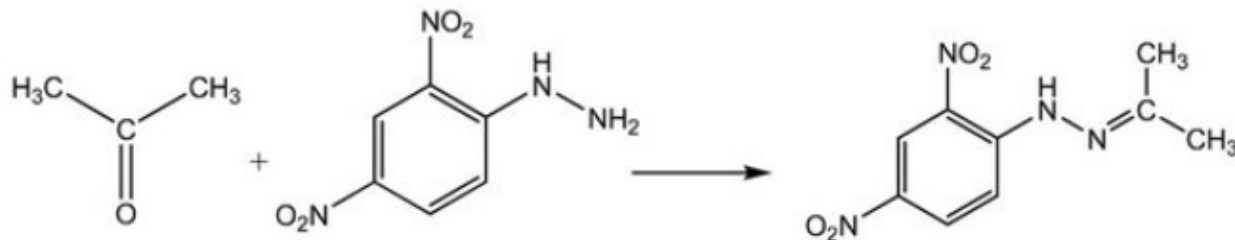
a) When benzene and bromine water were poured together, we have seen no color change. that indicate there was no reaction.

b) the solution formed from sulfuric acid, nitric acid and toluene was poured into a large test tube containing piece of ice. nitro benzene is formed as slight yellow oil.



5)KETONE

When DNPH and acetone were reacted, they formed noticeable precipitate.



6)PHENOL

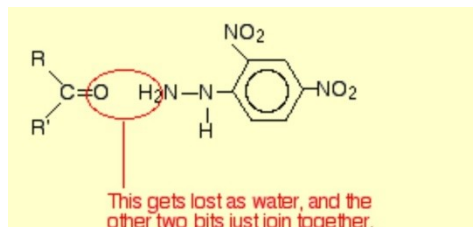
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When colorless phenol added to a color less neutral ferric chloride solution,resulted a purple final substance.

QUIZ QUESTIONS AND THEIR ANSWERS

1. Give the mechanism of the reaction of DNPH with acetone?

- In terms of mechanisms, this is a nucleophilic addition elimination reaction.the DNBH first adds across the carbon-oxygen double bond(the addition stage) to give an intermediate compound which then loses a molecule of water(the elimination stage).



2. What is the difference between addition and substitution reaction?

- Addition reaction occurs when 2 or more reactants combine to form a single products.this product will contain all the atoms that were present in a reactants.whereas the substitution reaction occurs when an exchange of elements in a reactant take place.atom or ion in the molecule is replaced by another atom, ion or group.

3. Give mechanism for the nitration of toluene?

- Electrophilic aromatic substitution represents an important class of reactions in organic synthesis. In “aromatic nitration” aromatic organic compounds are nitrated via an electrophilic aromatic substitution mechanism involving the attack of the electro-rich benzene ring on the nitronium ion.the formation is shown below.

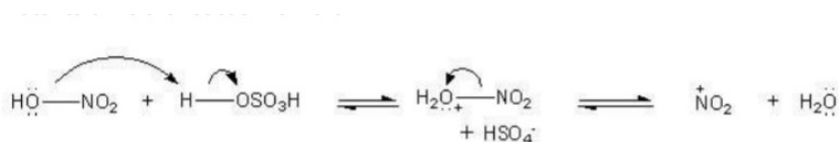


Figure 1: The mechanism for the formation of a nitronium ion.

- Toluene undergoes nitration to give ortho and para nitrotoluene isomers, but if it is heated it can give dinitrotoluene and ultimately the explosive trinitrotoluene(TNT).

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